

Multiple Choice

1. **Answer:** A

Options: A) 33.9; B) 29.8; C) 27.5; D) 31.2; E) 24.3

Note: The binding energy of the 4 He nucleus, calculated using the mass defect and Einstein's equation $E=mc^2$, yields approximately 33.9 MeV, reflecting the energy required to hold the nucleus together against the repulsive forces between protons.

2. **Answer:** A

Options: A) 0.4 henry; B) 0.3 henry; C) 0.2 henry; D) 0.6 henry; E) 0.7 henry

Note: The self-inductance of the coil can be calculated using the formula $L = \frac{V}{\frac{di}{dt}}$, where V is the back electromotive force (2 volts) and $\frac{di}{dt}$ is the rate of change of current (5 amperes per second), resulting in $L = \frac{2V}{5A/s} = 0.4 H$.

3. **Answer:** B

Options: A) It is increased by two times.; B) It remains the same.; C) It becomes zero.; D) It is reduced to half.; E) It is increased by eight times.

Note: The gravitational force between two objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between them, so doubling both the masses and the distance results in the force remaining unchanged, as both changes cancel each other out.

4. **Answer:** E

Options: A) (f / 3); B) (f / 2); C) (5 f / 3); D) (6 f / 5); E) (3 f / 4)

Note: The correct answer is derived from the lens formula for thin lenses in combination, where the effective focal length (F) of two lenses separated by a distance (d) is given by the formula $F = (f_1 * f_2) / (f_1 + f_2 - d)$ for lenses with focal lengths f_1 and f_2 , leading to $F = (f * f) / (f + f - (2f/3))$, which simplifies to $(3f/4)$.

5. **Answer:** A

Options: A) because Mars' axis is less tilted than the Earth's; B) because Mars has a smaller size than the Earth; C) because Mars has a slower rotation speed than the Earth; D) because Mars has more carbon dioxide in its atmosphere than the Earth; E) because Mars has a shorter year than the Earth

Note: The seasons in Mars' southern hemisphere are more extreme than on Earth because Mars has a smaller axial tilt (approximately 25 degrees compared to Earth's 23.5 degrees), resulting in greater variations in solar radiation received during its orbital path, which amplifies seasonal changes.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because a ratio of forward-bias to reverse-bias currents of -6×10^{25} indicates that the forward current is vastly greater than the reverse current, reflecting the exponential increase in carrier injection under forward-bias conditions compared to the negligible reverse current in a diode.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because an object with an angular size of 0.5 degrees and a height of 5 cm would actually be approximately 6 meters away from the observer, not 15 meters, based on the relationship between angular size, actual size, and distance.

8. **Answer:** False

Options: A) True; B) False

Note: A lens with a focal length of 20 inches cannot produce a virtual image when the object is placed 10 inches from it because the object is located within the focal length, which results in a real image rather than a virtual image.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because a thin film's absorption is not solely determined by its optical density but also by its thickness and refractive index, which together influence the fraction of incident light that is absorbed.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the Sun's surface temperature of approximately 6000 K places it in the G-type star category, which aligns with the established stellar temperature classification system that categorizes stars based on their effective temperatures.

Long Form Response

11. **Answer:** When 100 g of water at 0°C is mixed with 50 g of water at 50°C , the system reaches thermal equilibrium.

Note: correct because it correctly identifies the process of reaching thermal equilibrium, applies the conservation of energy principle, and uses the correct formula for heat transfer: $Q = mc\ln(T_{\text{final}}/T_{\text{initial}})$, where m is the mass, c is the specific heat capacity of water ($1 \text{ cal/g}^\circ\text{C}$), and T is the temperature.

11. **Answer:** To calculate the electrostatic force acting on a 10.0 g block with a charge

of $+8.00 \times 10^{-5} \text{ C}$ in the electric field $E = (3000i - 600j) \text{ N/C}$, we use the equation $F = qE$, where F is the force, q is the charge, and E is the electric field vector. Substituting the given values, we get $F = (8.00 \times 10^{-5} \text{ C}) * (3000i - 600j) \text{ N/C}$, which results in a force vector of $(0.24i - 0.048j) \text{ N}$. The magnitude of the force is $\sqrt{(0.24)^2 + (-0.048)^2} \text{ N} \approx 0.245 \text{ N}$.

Note: correct because it accurately applies the formula for electrostatic force, $F = qE$, using the given ch

End of Answer Key